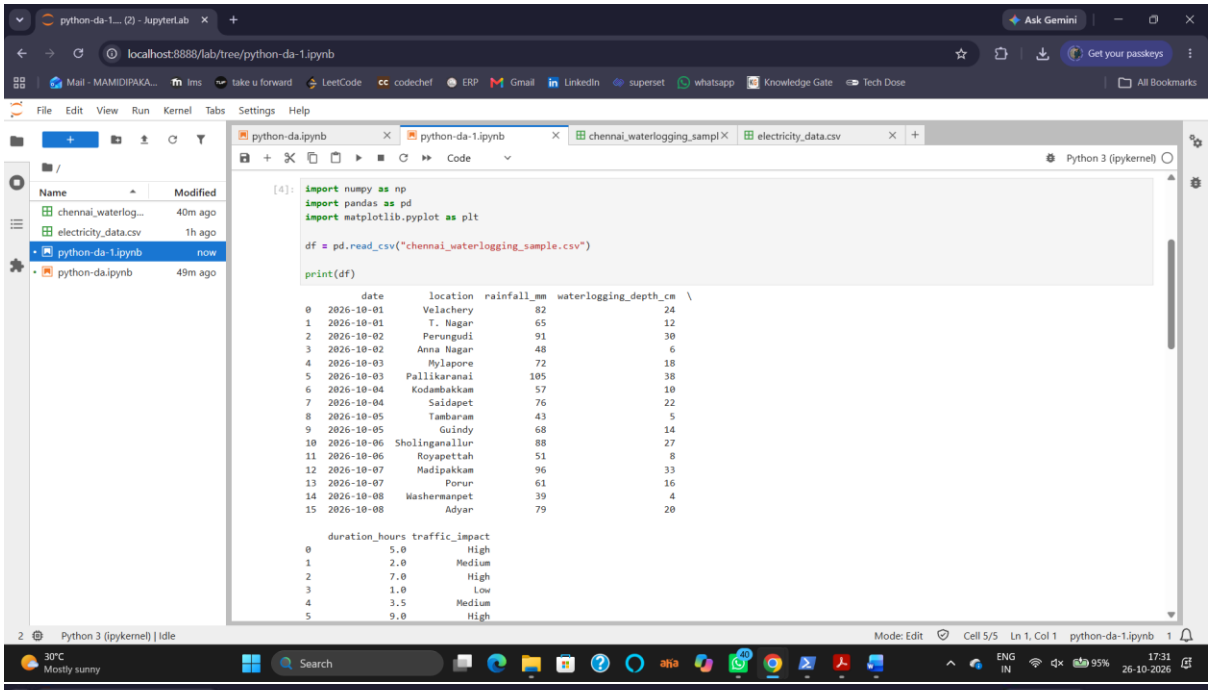


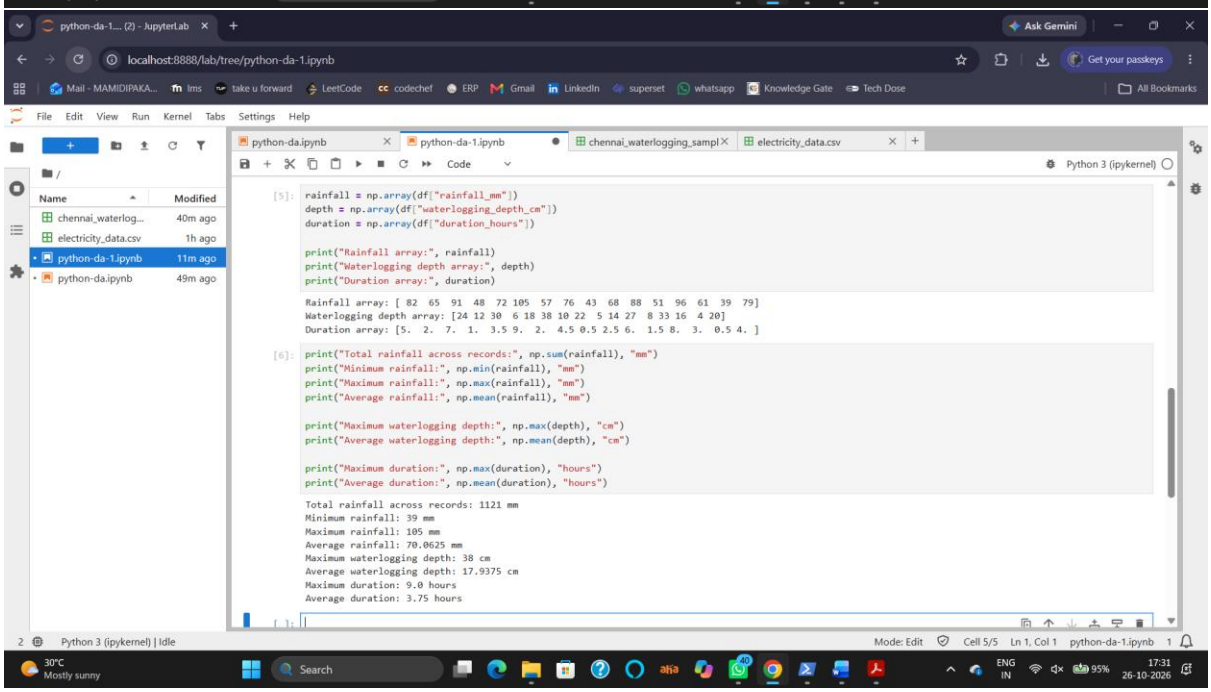
```
[4]:
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("chennai_waterlogging_sample.csv")

print(df)
```

```
         date       location  rainfall_mm  waterlogging_depth_cm  \
0  2026-10-01       Velachery           82                     24
1  2026-10-01        T. Nagar           65                     12
2  2026-10-02       Perungudi           91                     30
3  2026-10-02      Anna Nagar           48                      6
4  2026-10-03        Mylapore           72                     18
5  2026-10-03    Pallikaranai          105                     38
6  2026-10-04     Kodambakkam           57                     10
7  2026-10-04         Saidapet           76                     22
8  2026-10-05        Tambaram           43                      5
9  2026-10-05          Guindy           68                     14
10 2026-10-06  Sholinganallur           88                     27
11 2026-10-06      Royapettah           51                      8
12 2026-10-07       Madipakkam           96                     33
13 2026-10-07            Porur           61                     16
14 2026-10-08    Washermanpet           39                      4
15 2026-10-08           Adyar           79                     20

    duration_hours traffic_impact
0              5.0          High
1              2.0        Medium
2              7.0          High
3              1.0           Low
4              3.5        Medium
5              9.0          High
```



```
[5]:
rainfall = np.array(df["rainfall_mm"])
depth = np.array(df["waterlogging_depth_cm"])
duration = np.array(df["duration_hours"])

print("Rainfall array:", rainfall)
print("Waterlogging depth array:", depth)
print("Duration array:", duration)
```

```
Rainfall array: [ 82  65  91  48  72 105  57  76  43  68  88  51  96  61  39  79]
Waterlogging depth array: [24 12 30  6 18 38 10 22  5 14 27  8 33 16  4 20]
Duration array: [5.  2.  7.  1.  3.5 9.  2.  4.5 0.5 2.5 6.  1.5 8.  3.  0.5 4. ]
```

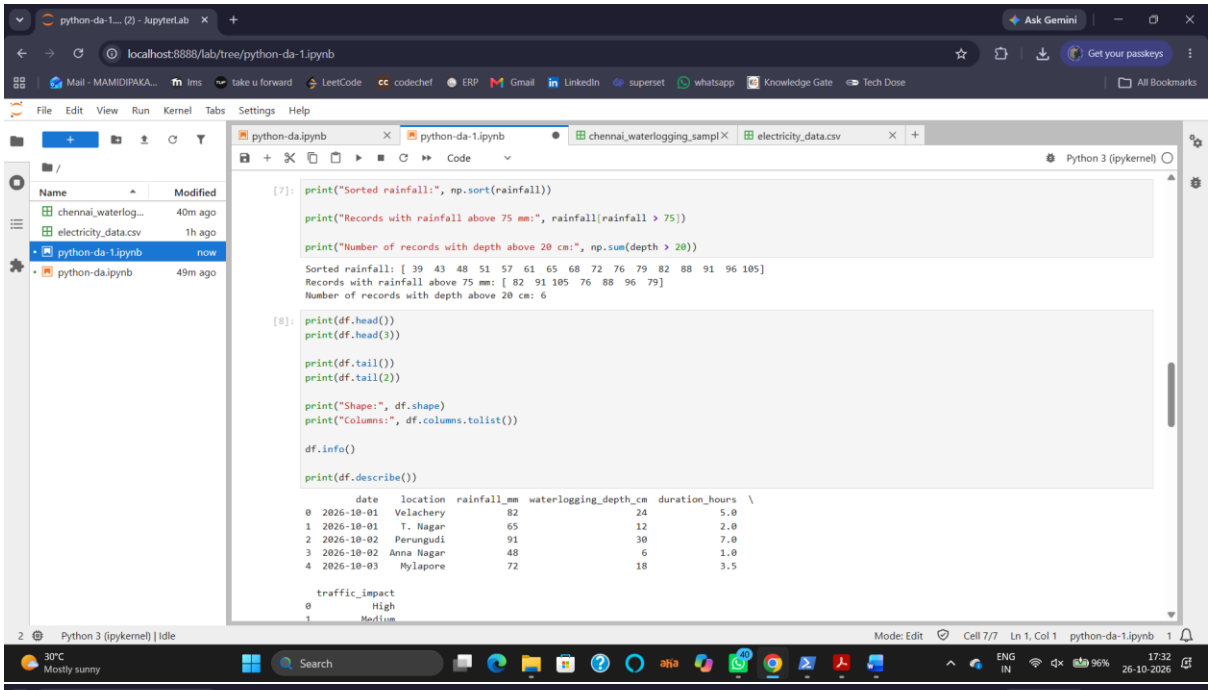
```
[6]:
print("Total rainfall across records:", np.sum(rainfall), "mm")
print("Minimum rainfall:", np.min(rainfall), "mm")
print("Maximum rainfall:", np.max(rainfall), "mm")
print("Average rainfall:", np.mean(rainfall), "mm")

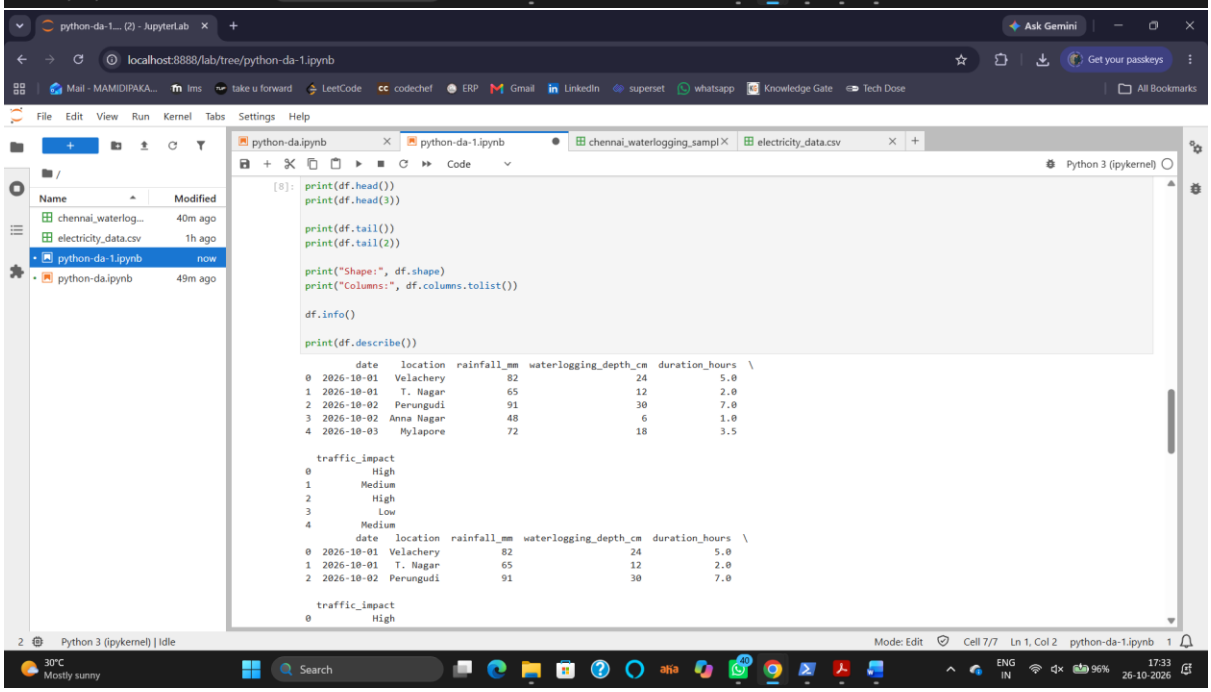
print("Maximum waterlogging depth:", np.max(depth), "cm")
print("Average waterlogging depth:", np.mean(depth), "cm")

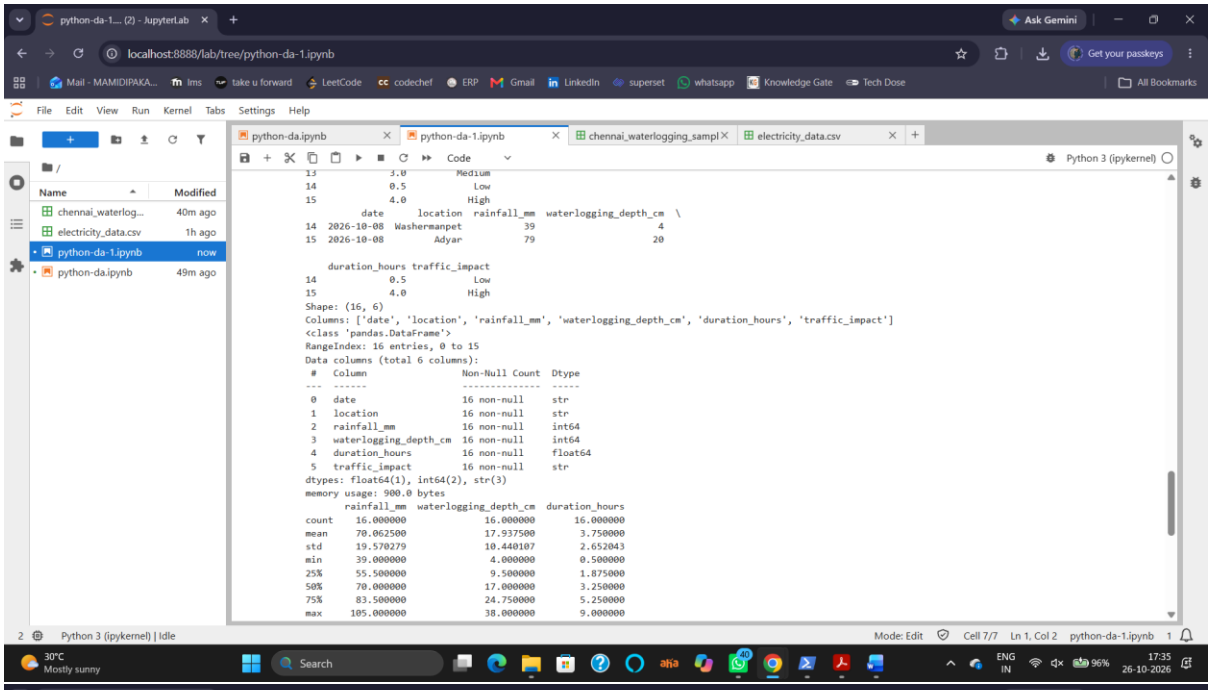
print("Maximum duration:", np.max(duration), "hours")
print("Average duration:", np.mean(duration), "hours")
```

```
Total rainfall across records: 1121 mm
Minimum rainfall: 39 mm
Maximum rainfall: 105 mm
Average rainfall: 70.0625 mm
Maximum waterlogging depth: 38 cm
Average waterlogging depth: 17.9375 cm
Maximum duration: 9.0 hours
Average duration: 3.75 hours
```

```
[ ]:
```

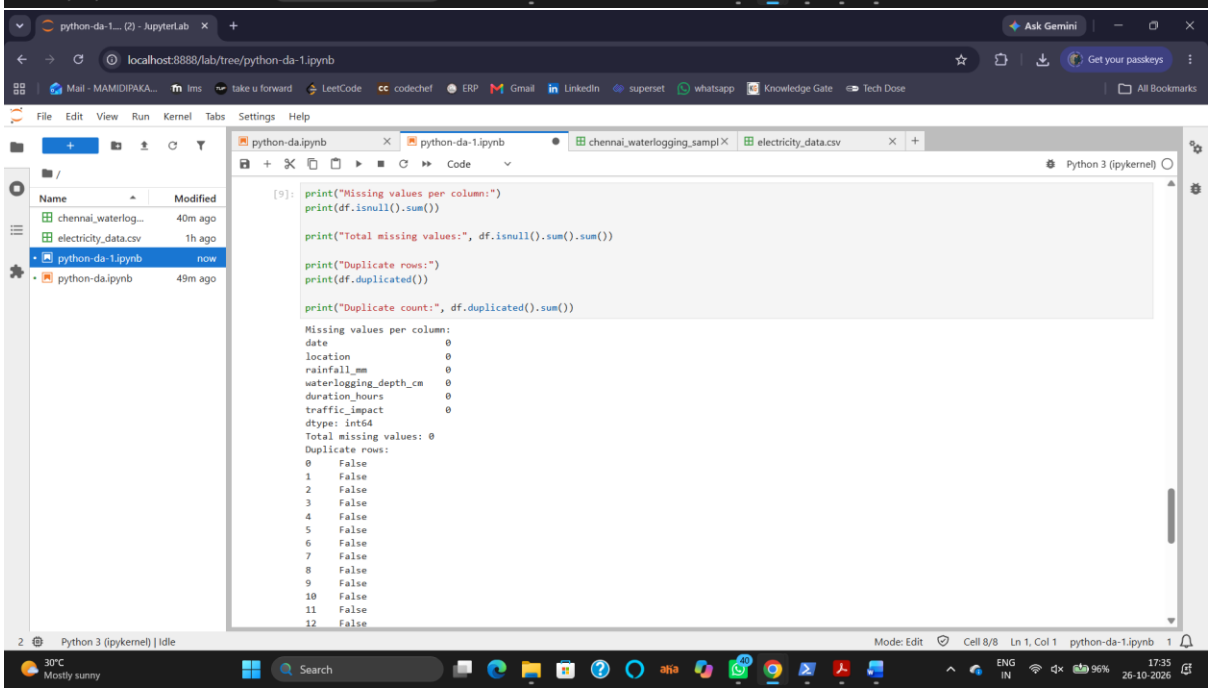

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localhost:8888/lab/tree/python-da-1.ipynb
Get your passkeys
File Edit View Run Kernel Tabs Settings Help
python-da.ipynb
python-da-1.ipynb
chennai_waterlogging_sampl
electricity_data.csv
Code
Python 3 (ipykernel)
Name
Modified
chennai_waterlog... 40m ago
electricity_data.csv 1h ago
python-da-1.ipynb now
python-da.ipynb 49m ago
[7]:
print("Sorted rainfall:", np.sort(rainfall))
print("Records with rainfall above 75 mm:", rainfall[rainfall > 75])
print("Number of records with depth above 20 cm:", np.sum(depth > 20))
Sorted rainfall: [39 43 48 51 57 61 65 68 72 76 79 82 88 91 96 105]
Records with rainfall above 75 mm: [82 91 105 76 88 96 79]
Number of records with depth above 20 cm: 6
[8]:
print(df.head())
print(df.head(3))
print(df.tail())
print(df.tail(2))
print("Shape:", df.shape)
print("Columns:", df.columns.tolist())
df.info()
print(df.describe())
date location rainfall_mm waterlogging_depth_cm duration_hours \
0 2026-10-01 Velachery 82 24 5.0
1 2026-10-01 T. Nagar 65 12 2.0
2 2026-10-02 Perungudi 91 30 7.0
3 2026-10-02 Anna Nagar 48 6 1.0
4 2026-10-03 Mylapore 72 18 3.5
traffic_impact
0 High
1 Medium
Python 3 (ipykernel) | Idle
Mode: Edit
Cell 7/7
Ln 1, Col 1
python-da-1.ipynb
30°C
Mostly sunny
Search
ENG
IN
96%
17:32
26-10-2026


python-da-1.... (2) - JupyterLab
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python-da.ipynb
python-da-1.ipynb
chennai_waterlogging_sampl
electricity_data.csv
Code
Python 3 (ipykernel)
Name
Modified
chennai_waterlog... 40m ago
electricity_data.csv 1h ago
python-da-1.ipynb now
python-da.ipynb 49m ago
[8]:
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print(df.head(3))
print(df.tail())
print(df.tail(2))
print("Shape:", df.shape)
print("Columns:", df.columns.tolist())
df.info()
print(df.describe())
date location rainfall_mm waterlogging_depth_cm duration_hours \
0 2026-10-01 Velachery 82 24 5.0
1 2026-10-01 T. Nagar 65 12 2.0
2 2026-10-02 Perungudi 91 30 7.0
3 2026-10-02 Anna Nagar 48 6 1.0
4 2026-10-03 Mylapore 72 18 3.5
traffic_impact
0 High
1 Medium
2 High
3 Low
4 Medium
date location rainfall_mm waterlogging_depth_cm duration_hours \
0 2026-10-01 Velachery 82 24 5.0
1 2026-10-01 T. Nagar 65 12 2.0
2 2026-10-02 Perungudi 91 30 7.0
traffic_impact
0 High
Python 3 (ipykernel) | Idle
Mode: Edit
Cell 7/7
Ln 1, Col 2
python-da-1.ipynb
30°C
Mostly sunny
Search
ENG
IN
96%
17:33
26-10-2026



```
13            3.0          Medium
14            0.5             Low
15            4.0            High
         date      location  rainfall_mm  waterlogging_depth_cm  \
14  2026-10-08  Washermanpet           39                      4
15  2026-10-08         Adyar           79                     20

    duration_hours traffic_impact
14             0.5             Low
15             4.0            High
Shape: (16, 6)
Columns: ['date', 'location', 'rainfall_mm', 'waterlogging_depth_cm', 'duration_hours', 'traffic_impact']
<class 'pandas.DataFrame'>
RangeIndex: 16 entries, 0 to 15
Data columns (total 6 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   date                  16 non-null     str
 1   location              16 non-null     str
 2   rainfall_mm           16 non-null     int64
 3   waterlogging_depth_cm 16 non-null     int64
 4   duration_hours        16 non-null     float64
 5   traffic_impact        16 non-null     str
dtypes: float64(1), int64(2), str(3)
memory usage: 900.0 bytes
       rainfall_mm  waterlogging_depth_cm  duration_hours
count    16.000000              16.000000       16.000000
mean     70.062500              17.937500        3.750000
std      19.570279              10.440107        2.652043
min      39.000000               4.000000        0.500000
25%      55.500000               9.500000        1.875000
50%      70.000000              17.000000        3.250000
75%      83.500000              24.750000        5.250000
max     105.000000              38.000000        9.000000
```



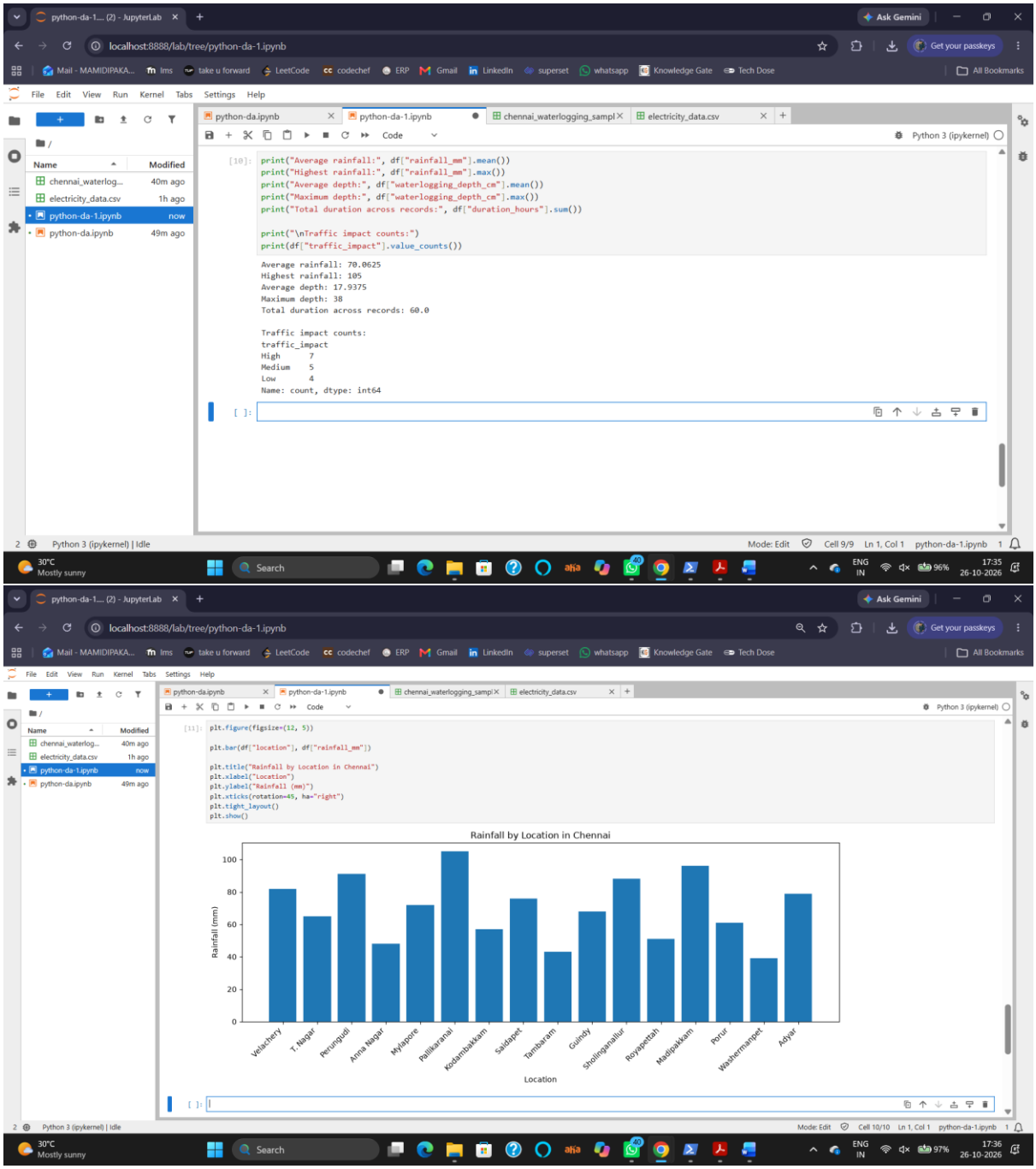
```
[9]: print("Missing values per column:")
     print(df.isnull().sum())

     print("Total missing values:", df.isnull().sum().sum())

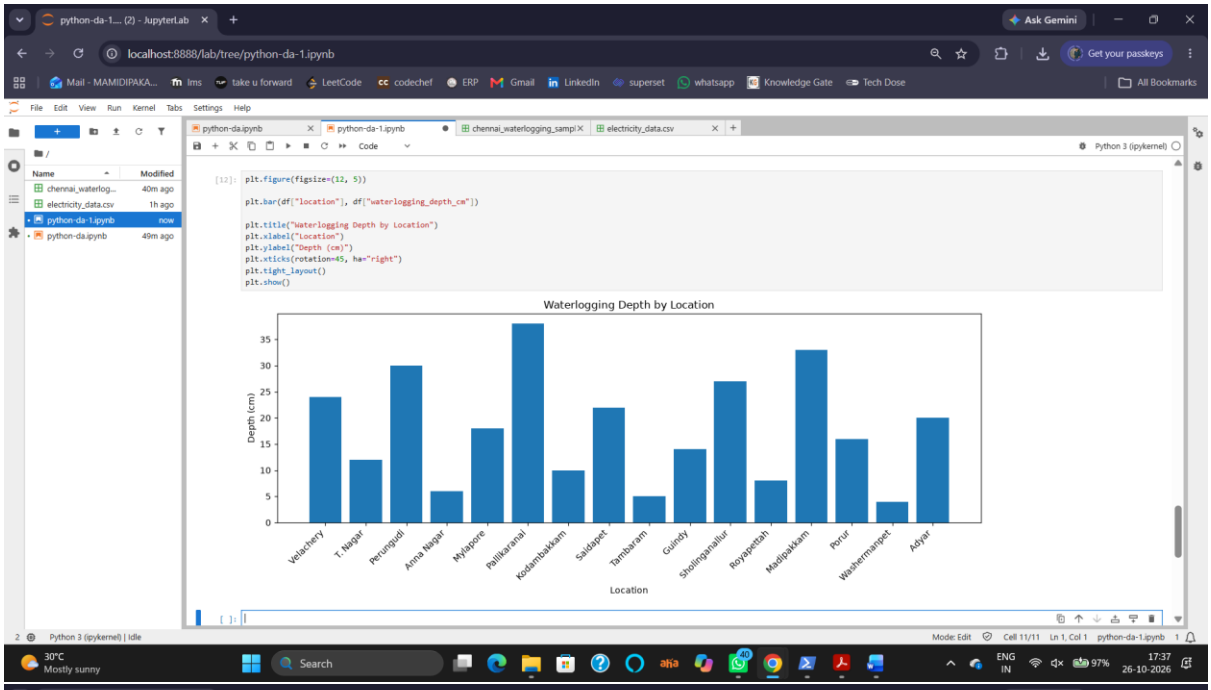
     print("Duplicate rows:")
     print(df.duplicated())

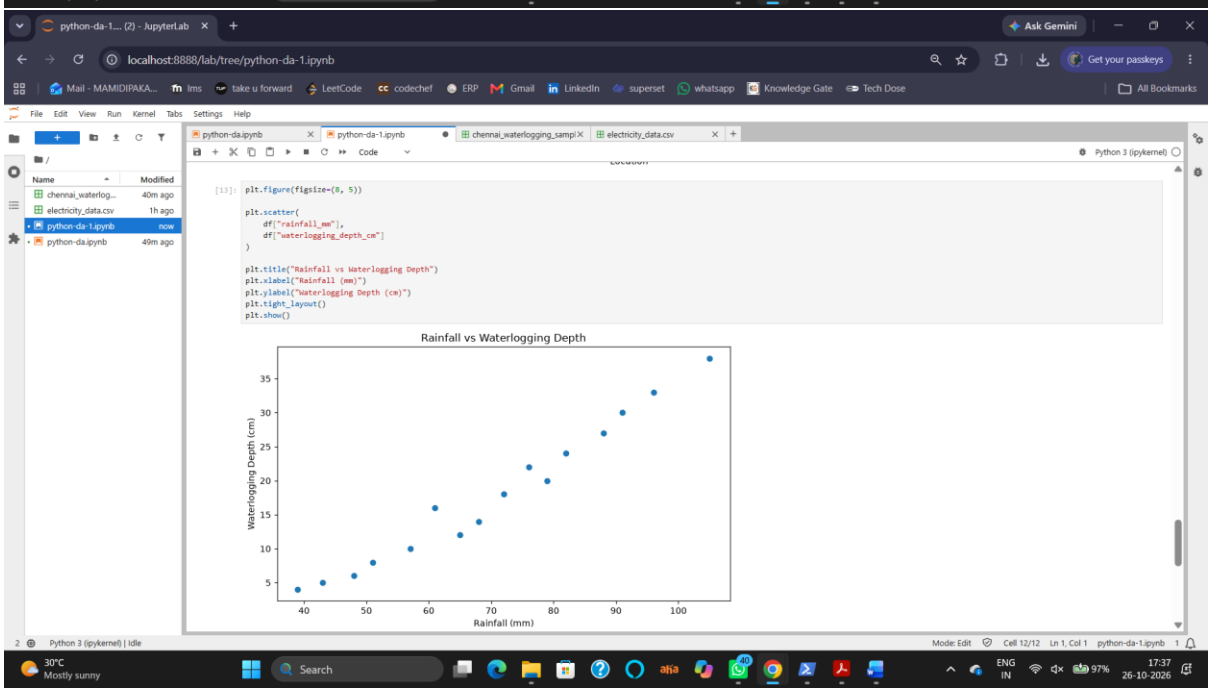
     print("Duplicate count:", df.duplicated().sum())

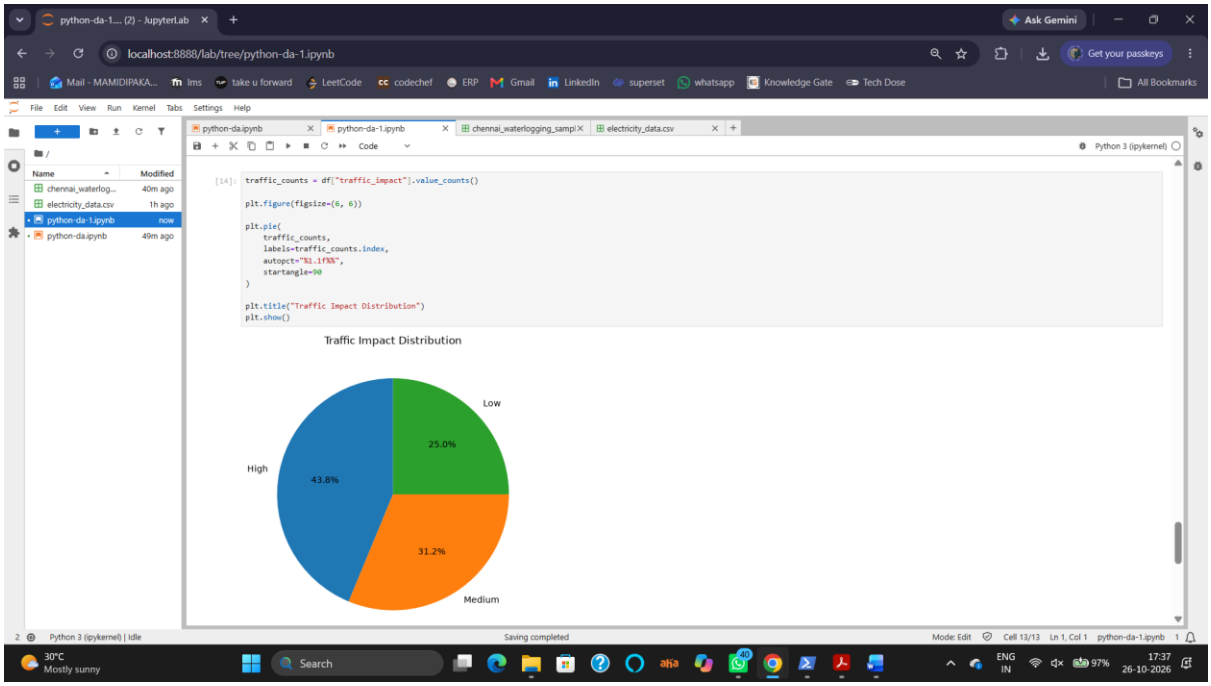
Missing values per column:
date                    0
location                0
rainfall_mm             0
waterlogging_depth_cm   0
duration_hours          0
traffic_impact          0
dtype: int64
Total missing values: 0
Duplicate rows:
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9     False
10    False
11    False
12    False
```


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File Edit View Run Kernel Tabs Settings Help
python-da.ipynb
python-da-1.ipynb
chennai_waterlogging_sampl
electricity_data.csv
Code
Python 3 (ipykernel)
Name
Modified
chennai_waterlog... 40m ago
electricity_data.csv 1h ago
python-da-1.ipynb now
python-da.ipynb 49m ago
[10]:
print("Average rainfall:", df["rainfall_mm"].mean())
print("Highest rainfall:", df["rainfall_mm"].max())
print("Average depth:", df["waterlogging_depth_cm"].mean())
print("Maximum depth:", df["waterlogging_depth_cm"].max())
print("Total duration across records:", df["duration_hours"].sum())
print("\nTraffic impact counts:")
print(df["traffic_impact"].value_counts())
Average rainfall: 70.0625
Highest rainfall: 105
Average depth: 17.9375
Maximum depth: 38
Total duration across records: 60.0
Traffic impact counts:
traffic_impact
High 7
Medium 5
Low 4
Name: count, dtype: int64
[ ]:
Python 3 (ipykernel) | Idle
Mode: Edit
Cell 9/9
Ln 1, Col 1
python-da-1.ipynb
30°C
Mostly sunny
Search
ENG
IN
96%
17:35
26-10-2026

python-da-1.... (2) - JupyterLab
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python-da.ipynb
python-da-1.ipynb
chennai_waterlogging_sampl
electricity_data.csv
Code
Python 3 (ipykernel)
Name
Modified
chennai_waterlog... 40m ago
electricity_data.csv 1h ago
python-da-1.ipynb now
python-da.ipynb 49m ago
[11]:
plt.figure(figsize=(12, 5))
plt.bar(df["location"], df["rainfall_mm"])
plt.title("Rainfall by Location in Chennai")
plt.xlabel("Location")
plt.ylabel("Rainfall (mm)")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.show()
Rainfall by Location in Chennai
Rainfall (mm)
0
20
40
60
80
100
Velachery
T. Nagar
Perungudi
Anna Nagar
Mylapore
Pallikaranai
Kodambakkam
Saidapet
Tambaram
Guindy
Sholinganallur
Royapettah
Madipakkam
Porur
Washermanpet
Adyar
Location
[ ]:
Python 3 (ipykernel) | Idle
Mode: Edit
Cell 10/10
Ln 1, Col 1
python-da-1.ipynb
30°C
Mostly sunny
Search
ENG
IN
97%
17:36
26-10-2026


python-da-1.... (2) - JupyterLab
localhost:8888/lab/tree/python-da-1.ipynb
Ask Gemini
Get your passkeys
File Edit View Run Kernel Tabs Settings Help
python-da.ipynb
python-da-1.ipynb
chennai_waterlogging_sampl
electricity_data.csv
Python 3 (ipykernel)
Name
Modified
chennai_waterlog... 40m ago
electricity_data.csv 1h ago
python-da-1.ipynb now
python-da.ipynb 49m ago
[12]:
plt.figure(figsize=(12, 5))
plt.bar(df["location"], df["waterlogging_depth_cm"])
plt.title("Waterlogging Depth by Location")
plt.xlabel("Location")
plt.ylabel("Depth (cm)")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.show()
Waterlogging Depth by Location
Depth (cm)
0
5
10
15
20
25
30
35
Velachery
T. Nagar
Perungudi
Anna Nagar
Mylapore
Pallikaranai
Kodambakkam
Saidapet
Tambaram
Guindy
Sholinganallur
Royapettah
Madipakkam
Porur
Washermanpet
Adyar
Location
[ ]:
Python 3 (ipykernel) | Idle
Mode: Edit
Cell 11/11
Ln 1, Col 1
python-da-1.ipynb
30°C
Mostly sunny
Search
ENG
IN
97%
17:37
26-10-2026


python-da-1.... (2) - JupyterLab
localhost:8888/lab/tree/python-da-1.ipynb
Ask Gemini
Get your passkeys
File Edit View Run Kernel Tabs Settings Help
python-da.ipynb
python-da-1.ipynb
chennai_waterlogging_sampl
electricity_data.csv
Python 3 (ipykernel)
Name
Modified
chennai_waterlog... 40m ago
electricity_data.csv 1h ago
python-da-1.ipynb now
python-da.ipynb 49m ago
[13]:
plt.figure(figsize=(8, 5))
plt.scatter(
df["rainfall_mm"],
df["waterlogging_depth_cm"]
)
plt.title("Rainfall vs Waterlogging Depth")
plt.xlabel("Rainfall (mm)")
plt.ylabel("Waterlogging Depth (cm)")
plt.tight_layout()
plt.show()
Rainfall vs Waterlogging Depth
Waterlogging Depth (cm)
5
10
15
20
25
30
35
40
50
60
70
80
90
100
Rainfall (mm)
Python 3 (ipykernel) | Idle
Mode: Edit
Cell 12/12
Ln 1, Col 1
python-da-1.ipynb
30°C
Mostly sunny
Search
ENG
IN
97%
17:37
26-10-2026


python-da-1.... (2) - JupyterLab
Ask Gemini
localhost:8888/lab/tree/python-da-1.ipynb
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File Edit View Run Kernel Tabs Settings Help
python-da.ipynb
python-da-1.ipynb
chennai_waterlogging_sampl
electricity_data.csv
Code
Python 3 (ipykernel)
Name
Modified
chennai_waterlog... 40m ago
electricity_data.csv 1h ago
python-da-1.ipynb now
python-da.ipynb 49m ago
[14]:
traffic_counts = df["traffic_impact"].value_counts()
plt.figure(figsize=(6, 6))
plt.pie(
traffic_counts,
labels=traffic_counts.index,
autopct="%1.1f%%",
startangle=90
)
plt.title("Traffic Impact Distribution")
plt.show()
Traffic Impact Distribution
Low
25.0%
High
43.8%
31.2%
Medium
Python 3 (ipykernel) | Idle
Saving completed
Mode: Edit
Cell 13/13
Ln 1, Col 1
python-da-1.ipynb
30°C
Mostly sunny
Search
ENG
IN
97%
17:37
26-10-2026